Rishabh Shah

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EDUCATION

Rutgers University, New Brunswick — **New Brunswick, NJ**
M.S. in Computer Science (GPA: 3.83/4) — *Sep 2025 - May 2027*

University of Mumbai — **Mumbai, India**
B.Tech. in Artificial Intelligence and Data Science (CGPA: 7.76/10) — *Dec 2021 - Jun 2025*

SKILLS

Languages: Python, Java, C++, C, TypeScript, JavaScript, SQL
Backend: Django, Django ORM, Express.js, REST APIs, JWT, OAuth2, RBAC, Zod, Prisma
Frontend: React, Next.js, Tailwind CSS, HTML, CSS
Databases: PostgreSQL, MySQL, MongoDB, SQLite, Redis
Cloud and DevOps: AWS (EC2, S3), Google Cloud Platform, Azure, Vercel, Docker, CI/CD, GitHub Actions, Git, Linux
Testing: Pytest, Vitest, Playwright, React Testing Library, Postman

EXPERIENCE

Teaching Assistant — **Sep 2026 - Present**
Rutgers University, CS205: Introduction to Discrete Structures I (Prof. Samaneh Hamidi) — *New Brunswick, NJ*

- Led weekly recitations for 50+ undergraduates on set notation, propositional logic, truth tables, and Boolean circuits.
- Taught first-order logic, predicates, quantifiers, and induction proofs for program correctness, trees, and grammars.
- Covered relations and closures, orders, equivalence relations, functions, and finite-state machines with CS applications.

Software Engineering Intern — **Jan 2025 - Jul 2025**
Aaditya Technologies — *Mumbai, India*

- Reduced P95 API latency by 86% (3,100 ms to 420 ms) by profiling Django request paths, optimizing ORM-generated MySQL queries, adding targeted indexes, tuning connection pooling, and trimming response payloads.
- Reduced recurring API integration defects by 30% across 12+ REST endpoints by standardizing Django serializer validation, request and response schemas, error handling, and HTTP status codes.
- Secured 4 Django backend components by implementing JWT authentication and RBAC for admin, staff, and client roles.
- Reduced deployment setup effort by 65% for a 4-person team by containerizing Django services with Docker and standardizing AWS EC2, S3, and environment configuration.

Web Developer Intern — **Jun 2024 - Sep 2024**
Creative Line — *Mumbai, India*

- Reduced P95 API latency by 50% (620 ms to 310 ms) across 8 endpoints by optimizing Express.js handlers, indexing MySQL queries, and eliminating redundant database calls.
- Shipped 4 customer-facing features across 10+ REST API endpoints by building Next.js interfaces, Express.js services, and validated MySQL CRUD workflows.
- Achieved zero rollbacks across 9 production deployments by writing 55+ automated tests and enforcing regression testing, defect fixes, and pre-release validation.

PROJECT

CseGraph: Code Context Retrieval Engine — ***GitHub* | *PyPI***

- Reduced median repository context by 82.3% (1,540 to 273 tokens) across 364 tasks by combining FTS5/BM25 search, symbol resolution, graph ranking, and token-bounded retrieval.
- Indexed 177K+ symbols across 12,245 files by building a Tree-sitter pipeline for Python, JavaScript, and TypeScript that stores code relationships in a transactional SQLite graph.
- Ensured safe incremental re-indexing, verified by 379 passing tests at 76% coverage, by implementing SHA-256 change detection and atomic index replacement across 9 CLI commands and 6 MCP tools.

PUBLICATIONS AND ACHIEVEMENTS

- **Intelligent Traffic Surveillance**: IEEE ICICIS 2025, Scopus indexed.
- **Bridging Financial Data Gaps with WGAN-GP: Generating Synthetic Time Series for Robust Models**: Book Chapter, *Explainable Artificial Intelligence: Bridging Concepts, Applications, and Future Challenges*, Scrivener Publishing, 2026.
- **JARVIS: Voice Assistant with Smart Home Automation**: ISTE 2025.
- **Winner, Rutgers Health Hack 2026**: Won the Sevaro track for Cerevia, a real-time tele-neurology command center that forecasts staffing gaps and recommends credential-aware physician shifts.
- **Winner, DivHacks 2026**: Won MLH Best Use of Backboard.io at Columbia University for PlaNYt, an AI NYC trip planner.